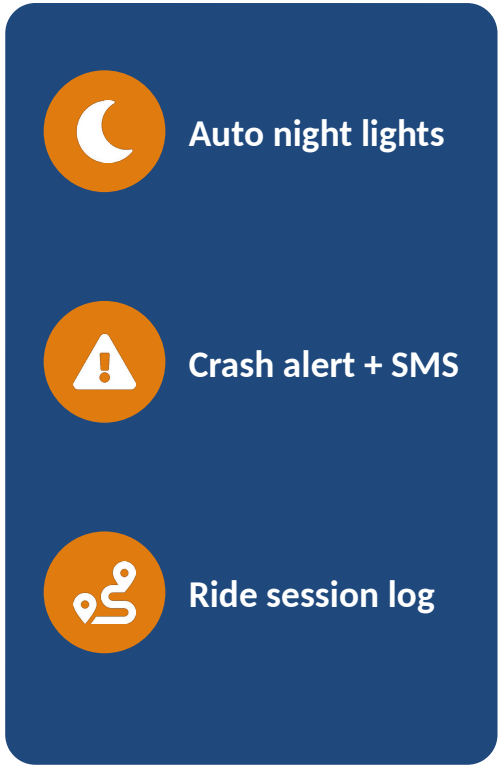
Cover Page

Problem Statement/ Concept Title	**Smart Helmet Light: auto lights, crash alert and ride log for two-wheeler riders**
Theme	*[Theme as registered on portal]*
Team Name (Registered on portal)	*[Team Name]*
Team ID (Registered on portal)	*[Team ID]*
State/UT	*[State/UT]*



[Team Name]

Problem – Proposed Solution

Hack for Social Cause

CORE PROBLEM

≈9 two-wheeler rider deaths every hour in India (2023)

45% of all road deaths are two-wheeler riders (2023)

69% of road deaths happen in rural India (2023)

Source: MoRTH data, analysis in ref. [2]

PROPOSED SOLUTION: ONE ESP32 HELMET + THE RIDER'S PHONE

Auto night mode

LDR senses dusk; front warm-white and rear red lights switch on

Fixes: Low visibility

Crash alert

Impact + stillness → 10 s buzzer → SMS with GPS to contact

Fixes: Nobody calls for help

Session tracking

BLE to phone: route, duration, distance, hard braking

Fixes: No ride record

INNOVATION AND UNIQUENESS

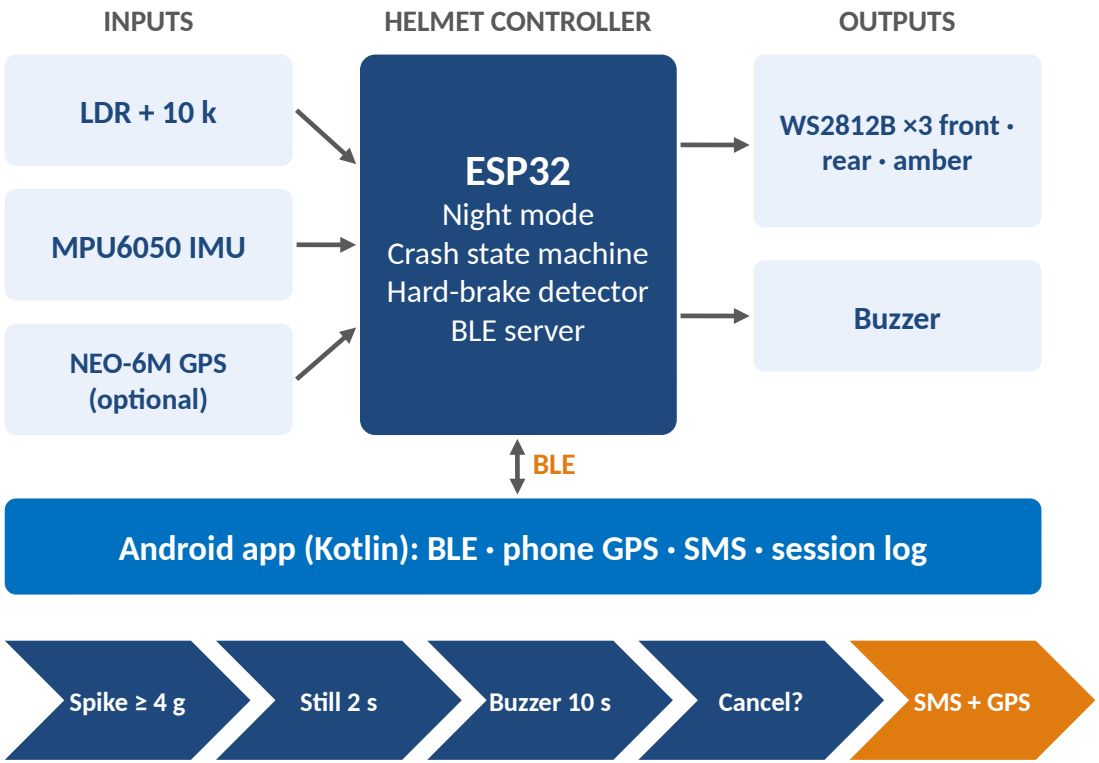
Phone provides GPS + SMS: no SIM or GPS module in the helmet

10 s cancel window and stillness check cut false alarms

Indicative parts cost ₹1,500–2,000; open-source

[Team Name]

Technical Approach



Crash flow

TECHNOLOGIES

C++ / Arduino-ESP32 (PlatformIO)
Kotlin, Android BLE + SmsManager
JSON over BLE GATT

GITHUB / GITLAB

github.com/[user]/smart-helmet-light

DEMO VIDEO

[video link]

EXTENT OF AI TOOLS

Claude: code scaffolding, docs, slides. Team: wiring, testing, review [edit to match]

[Team Name]

Impact Potential

Impact on target audience

- Daily commuters and night riders
- Delivery and gig riders
- Rural and highway riders

45% of road deaths are two-wheeler riders (2023) [1][2]

Benefits (social, economic)

Social

- Visible at dusk and night
- Help alerted automatically
- Ride data for safer habits

Economic

- Indicative ₹1,500–2,000
- Reuses the rider's phone

Stakeholders

- Riders
- Families / emergency contacts
- Delivery fleets and employers
- Road-safety bodies, traffic police
- Helmet makers

[Team Name]

Feasibility and Risk Analysis

FEASIBILITY

- Off-the-shelf, low-cost parts
- Detector logic tested on a PC: 8 of 8 scenarios pass
- Open-source stack, MIT licence
- Works with any Android 8.0+ phone

Status: [prototype stage – edit]

Potential challenge / risk	Strategy to overcome it
False alarms (potholes, drops)	4 g spike + stillness check; 10 s cancel window
BLE link drops after a crash	Phone timer still sends the SMS
No phone, signal or SMS credit	Buzzer + amber lights still warn; limit stated
Thresholds need field tuning	All values in config.h; tune with replay traces
Battery and LED current	Brightness capped; protected TP4056 + Li-ion

[Team Name]

Team

TEAM COMPOSITION AND ROLES

[Member name]

Embedded firmware

ESP32, sensors, LEDs

[Member name]

Phone app

BLE, GPS, SMS, logging

[Member name]

Hardware

Wiring, power, mounting

[Member name]

Research and pitch

Data, deck, demo

CHALLENGES

- Telling a crash from a pothole
- Keeping the SMS reliable if BLE drops
- Powering LEDs and ESP32 from one cell

OUR JOURNEY

[Add 2–3 short lines: how the idea started, tests done, what you learned]

[Team Name]

References

[1] Ministry of Road Transport and Highways, Road Accidents in India 2023 (Transport Research Wing), morth.nic.in

[2] Newslaundry, “More two-wheelers, gig workers under pressure: In India, 9 bike riders die every hour”, 10 Nov 2025 (analysis of MoRTH data), newslaundry.com

[3] InvenSense (TDK), MPU-6000 / MPU-6050 Product Specification and Register Map

[4] Espressif, Arduino-ESP32 documentation (BLE library), docs.espressif.com/projects/arduino-esp32

[5] Adafruit, NeoPixel Überguide (WS2812B power and wiring), learn.adafruit.com/adafruit-neopixel-uberguide

[6] NanJing Top Power, TP4056 1 A Standalone Linear Li-Ion Battery Charger datasheet

[7] Android Developers: Bluetooth LE overview, SmsManager, Foreground services, developer.android.com

[8] *Project repository: code, docs, sample data (synthetic), github.com/[user]/smart-helmet-light*

Sample data in the repository is synthetic. Thresholds are starting values pending field tests.